Maximum Likelihood Estimator (MLE)

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1 Beyond MM Estimators

Consider the previous example of the exponential distributed random variable, which we have seen last time. We could have used two moments, instead of only one, to estimate the parameter β. More generally, we could use only d moments to estimate a d dimensional parameter vector, but by dismissing the $q - d > 0$ additional moments, we would loose information. The remedy for this situation was introduced in the econometrics literature by Hansen (1982) in his famous article, and it is called the *Generalized Method of Moments* (GMM). The idea behind GMM estimation is that even though it is impossible to solve q system of equations provided by the sample moment conditions to estimate $d < q$ parameters, we can still find an estimate of the parameter vector θ that brings the sample moments as close as possible to the population moments. We estimate the parameter vector by minimizing the sum of squares of the differences between the population moments and the sample moments. This yields the minimum variance estimator in the class of estimators that use these moment conditions.

2 Maximum Likelihood Estimator

The MLE estimator is a point estimator, like the method of moments. It was first proposed by the geneticist/statistician Ronald A. Fisher in 1922. The basic idea is to find the values of the parameters which are more plausible to produce the data that we observe. The method assumes that *we know* the structural form of the probability density function of the samples, but *we do not know* the parameters of this density function.

Definition 2.1. *Let $f(x_1, x_2, \ldots, x_n; \theta)$, where $\theta \in \Theta \subseteq \mathbb{R}^k$, be the joint probability density function of n random variables $X_1, \ldots, X_n$ with sample values $x_1, \ldots, x_n$. The* ***likelihood function*** *of the sample is given by*

$$L(\theta | x_1, x_2, \ldots, x_n) = f(x_1, x_2, \ldots, x_n; \theta).$$

Hence, the likelihood of $\theta = 1$ is the probability of observing $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$ when $\theta = 1$. More generally, the likelihood of θ given an observed sample $x = (x_1, \ldots, x_n)$, is the probability of a random sample taking value x under the parameter θ. If we fix θ, the likelihood is the probability density function of the random variable X. The maximum likelihood estimator

(MLE) of θ is defined as the maximizer of the likelihood, over θ.

Example 1: Consider the estimation of the mean μ and variance σ^2 of a Gaussian distribution from i.i.d. sample observations $x_1, x_2, \ldots, x_n$. The parameter vector is $\theta := (\mu, \sigma)$. The likelihood function is

$$L(\theta|x) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

To find the MLE estimates of μ and σ, we need to maximize $L(x\big|\theta)$ over μ and σ. This is equivalent to maximizing the log-likelihood function, given by

$$\log\left(L(\theta|x)\right) = -n\log\left(\sqrt{2\pi}\right) - \frac{n}{2}\log\left(\sigma^2\right) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}$$

We can then find the values of μ and σ that maximize the log likelihood by taking first-order derivatives and setting them equal to zero. By doing so, we obtain

$$\partial_\mu \sum_{i=1}^{n}(x_i-\mu)^2 = 0 \leftrightarrow \hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

$$-\frac{n}{2}\partial_{\sigma^2}\log(\sigma^2) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2}\partial_{\sigma^2}\frac{1}{\sigma^2} = 0 \leftrightarrow \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2 \tag{2}$$

Notice that while the MLE estimate of the mean corresponds to the sample mean, the MLE estimate of the variance differs from the sample variance estimator. Indeed, in the MLE estimate of the variance we divide by n, as opposed to $n-1$. This also means that the MLE produces a biased estimate of the variance. Obviously, this matters when n is small. If n is sufficiently large, then it matters much less whether we divide by n or $n-1$, and both estimators are asymptotically equivalent.

Example 2: Let $X_1, X_2, \ldots, X_n$ be a random sample from the uniform distribution on the interval $(0, \theta)$, where $\theta > 0$. We want to find the MLE of θ. Notice that the probability density function of the uniform is

$$f(x) = \begin{cases} 1/\theta & \text{if } 0 \le x \le \theta \\ 0 & \text{otherwise} \end{cases}.$$

Hence, the likelihood function is given by

$$L(\theta|x) = \begin{cases} 1/\theta^n & \text{if } 0 \le x_1, x_2, \ldots, x_n \le \theta \\ 0 & \text{otherwise} \end{cases}.$$

Let us sort the data from smallest to largest, i.e. let $x_{(1)} \le x_{(2)} \cdots \le x_{(n)}$ be the order statistics. Then, taking the derivative of the log-likelihood function with respect to θ (recall that maximizing the log-likelihood is equivalent to maximizing the likelihood function), we obtain

$$\partial_\theta \log L(\theta|x) = -\frac{n}{\theta} < 0.$$

Hence, the log-likelihood function (and thus the likelihood function) is a decreasing function of θ. Consequently, it is maximized at $\hat{\theta}_{MLE} = x_{(n)}$.

In general, the procedure to find MLE consists of the following steps:

- Define the likelihood function $L(\theta|x)$
- Take the natural logarithm of $L(\theta|x)$ (often easier to work with the log as it transforms a product into a sum, which is easier to differentiate)
- When possible, differentiate $\log L(\theta|x)$ and then equate the derivative to zero
- Solve for the parameter θ, and obtain $\hat{\theta}_{MLE}$
- Check that this is a global maximizer, by verifying that the log-likelihood function is concave in θ.

Things may be more difficult. For instance, a closed form expression for $\hat{\theta}_{MLE}$ may not be obtained by analytically differentiating the log-likelihood function, or there may exist multiple local optima, and it may be hard to compare them. In this case, one needs to resort to numerical methods to find the maximum likelihood estimates. These methods include Newton's method, quasi-Newton method, direct search method, etc... They can be implemented by the R functions optimize(), optim().

Pro and Cons of the MLE: The pros are the following. It uses information about the statistical distribution which generates the data. We will discuss next that when the sample size n is large enough, it is unbiased, consistent, normally distributed, and efficient. Efficiency means that it produces the minimum MSE among all methods, including the methods of moments. The cons are the following. MLE can be highly biased for small samples. In many cases, the MLE has no closed-form solution. In this case, numerical methods need to be employed. These methods are sensitive to the starting values in the optimization, and might not give a global optimum. This problem is especially common when θ is high dimensional.